NCUR Abstract

Abstract Title: An Empirical Study of Predictive Uncertainty Under the Right to Be Forgotten

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Abstract Text: Privacy regulations such as the Right to be Forgotten compel AI systems to delete specific private data upon request. To comply, developers use machine unlearning, a technique that removes data without having to retrain the model from scratch. However, it is unclear if this process leaves the model reliable. A trustworthy model requires calibrated predictive uncertainty; effectively, if a system claims 90% confidence in a diagnosis, it must be correct 90% of the time.

This study empirically evaluates whether unlearning compromises that calibration. We utilized a Convolutional Neural Network (CNN) backbone on two datasets: CIFAR-10 for benchmarking and DermaMNIST to simulate clinical medical imaging. We tested three distinct unlearning mechanisms - SISA, First-Order updates, and Fine-Tuning, while removing data fractions ranging from 5% to 20%. To assess the impact, we employed Uncertainty Quantification techniques, including Deep Ensembles and Monte Carlo Dropout, measuring shifts in Expected Calibration Error and Brier Score.

We anticipate that aggressive unlearning strategies will degrade model calibration more severely than standard training, resulting in overconfident predictions. Furthermore, we examine whether removing entire classes of data causes greater uncertainty drift than removing random instances. This research aims to identify which unlearning methods preserve safety for real-world deployment.